DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer 2024 **Course: B. Tech.**

Branch: Artificial Intelligence and Data Science Engineering and Allied

Semester VI **Subject Code & Name:** BTAIC601 Deep Learning

Max Marks: 60 **Date: 13/06/2024** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Write proper Syntax, example and program wherever necessary.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Explain Multilayer Perceptron.	**Remember**	**6**
B)	What is Gradient Descent and its variants.	**Understanding**	**6**
C)	Differentiate between forward propagation and backpropagation networks.	**Synthesis**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	Explain Building Blocks of CNN.	**Synthesis**	**6**
B)	Differentiate Fully Connected Network vs Convolutional Neural Network.	**Understanding**	**6**
C)	Give different CNN Architectures and explain any two in detail.	**Analysis**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	Explain Transfer learning in vision (image data).	**Analysis**	6
B)	Discuss Advantages & Disadvantages of transfer learning. Give uses.	**Synthesis**	6
C)	Explain Regularization.	**Understanding**	6
Q.4	**Solve Any Two of the following.**		**12**
A)	Give brief introduction to Sequential/Temporal Data.	**Remember**	**6**
B)	Explain working of Recurrent Neural Network (RNN).	**Synthesis**	**6**
C)	Explain types of Recurrent Neural Networks (RNNS).	**Understanding**	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	Explain Variational Autoencoder.	**Remember**	**6**
B)	Explain Generative Adversarial Network (GAN) architecture.	**Analysis**	**6**
C)	Explain applications of Deep Learning algorithms.	**Understanding**	**6**

***** End *****